Band-1 Formal-Conjectures Stub Probe: Erdős 742 (Murty–Simon) Finite Check

Human + ZoraASI full-dash (executor)

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Honesty stamp. This note does **not** claim that Erdős Problem 742 is solved. It does **not** stamp Clay, and it does **not** claim Erdős problems solved en masse. One concrete item only: Formal Conjectures Lean stub `ErdosProblems/742.lean`, plus an independent finite computational check for $n \leq 7$.

Abstract

We execute tonight's Band-1 collapse item 2: pick one Formal-Conjectures Erdős item that is finite-check and already has a Lean stub; close or refute that stub — or document why a Lean close is unavailable and do a finite check with honesty.

Chosen item: Erdős Problem 742 (Murty–Simon conjecture on diameter-2-critical graphs), formalized at

`https://github.com/google-deepmind/formal-conjectures/blob/main/FormalConjectures/ErdosProblems/742.lean`

Lean status on this box: `lean` / `elan` / `lake` are **not** installed (`which lean` exits nonzero). Therefore the stub cannot be closed or refuted *inside Lean* tonight. We document the stub contents and state an explicit Lean target, then run a pure-Python exhaustive finite check for labeled graphs on $n = 3, \ldots, 7$.

Finite-check result: for every $n \in \{3, 4, 5, 6, 7\}$, every diameter-2-critical graph on n labeled vertices has at most $\lfloor n^2/4 \rfloor$ edges (0 violators). Balanced complete bipartite graphs $K_{\lfloor n/2 \rfloor, \lceil n/2 \rceil}$ are diameter-2-critical and attain the bound. Wall time for $n = 7$ enumeration: ≈ 43.4 s.

Problem statement (informal)

A simple undirected graph G is **diameter-2-critical** if $\operatorname{diam}(G) = 2$ and deleting any edge increases the diameter (equivalently $\operatorname{diam}(G - e) \neq 2$).

Murty–Simon conjecture (Erdős 742). If G is diameter-2-critical on n vertices, then

$$|E(G)| \leq \left\lfloor \frac{n^2}{4} \right\rfloor,$$

with equality for the balanced complete bipartite graph $K_{\lceil n/2 \rceil, \lfloor n/2 \rfloor}$.

Why finite-check? Fan verified the conjecture for all $n \leq 24$ and $n = 26$; Füredi proved it for all sufficiently large n. The remaining gap is a finite (but large) verification problem — hence the Formal Conjectures page and teorth/erdosproblems classify the item as decidable / open-but-finite.

Lean stub (discovered)

File: `FormalConjectures/ErdosProblems/742.lean` (google-deepmind/formal-conjectures, main).

Headline (still open; answer placeholder):

```
@[category research open, AMS 5]
theorem erdos_742 :
    answer(sorry)    (V : Type*) [Fintype V] [DecidableEq V]
    (G : SimpleGraph V) [DecidableRel G.Adj], IsDiameter2Critical G →
    G.edgeFinset.card  (Fintype.card V) ^ 2 / 4 := by
  sorry
```

Solved variants that remain `sorry` in-repo (literature proofs, not Lean-closed here):

- `erdos_742.variants.plesnik_bound` — $|E| < 3n(n-1)/8$
- `erdos_742.variants.fan_bound` — bound for $n \leq 24$ or $n = 26$
- `erdos_742.variants.furedi_bound` — sufficiently large n

Proved-in-file sanity check (no `sorry`): `complete_bipartite_edge_count` — $|E(K_{a,b})| = a \cdot b$.

Why the stub is not closed tonight

Check	Result
Stub discoverable on GitHub	**Yes** — `742.lean`
Finite / decidable character	**Yes** — Fan + Füredi leave a finite gap
`lean` / `elan` / `lake` on box	**Absent**
Mathlib / lake project cloned	**No** (would require elan + large download)
Lean close/refute of `answer(sorry)`	**Not performed**

Closing `erdos_742` with `answer(True)` would claim the full conjecture for **all** n, which tonight's $n \leq 7$ check does **not** justify. Honest Lean targets (below) are small-n specializations of `fan_bound`, not a mass Erdős stamp.

Lean targets (for a future Lean session)

1. **Immediate:** prove a specialization `erdos_742.variants.n_le_7` stating the Murty–Simon inequality for `Fintype.card V  7`, matching tonight's exhaustive certificate (or import a verified enumeration certificate).
2. **Literature bridge:** replace `sorry` in `erdos_742.variants.fan_bound` by a formal citation/proof for Fan's $n \leq 24$, $n = 26$ cases.
3. **Not tonight:** do **not** replace `answer(sorry)` by `answer(True)` in `erdos_742` without a complete proof covering the Fan–Füredi gap.

Method (finite computational check)

Represent a labeled simple graph on n vertices by a bit-mask on the $\binom{n}{2}$ possible edges. For each mask:

1. Build adjacency lists.
2. Compute diameter by BFS from every vertex (disconnected $\mapsto +\infty$).
3. Keep only graphs with diam $= 2$.
4. For each edge e, recompute diameter of $G - e$; require $\mathrm{diam}(G - e) \neq 2$.
5. Record $|E|$; flag any critical graph with $|E| > \lfloor n^2/4 \rfloor$.

Additionally, for each $n = 3..7$, construct $K_{\lfloor n/2 \rfloor, \lceil n/2 \rceil}$ and verify it is diameter-2-critical and attains the bound.

Implementation: pure Python 3 (`/workspace/mission-control/labs/erdos_742_murty_simon/verify_murty_simon_sma`
No SAT solver; no Lean.

Experiment design

Parameter	Choice
Probe	Formal-Conjectures Erdős stub + finite check
Slug	`erdos_742_murty_simon`
Host	Linux box, Python 3; Lean absent
Search	Exhaustive labeled enumeration, $n = 3..7$
Success criterion	0 violators of $\lfloor n^2/4 \rfloor$; bipartite witnesses OK; document Lean gap
Overclaim ban	No full Murty–Simon proof; no Clay; one problem only

Results (numbers!)

Exhaustive labeled census

n	Graphs $2^{\binom{n}{2}}$	Diam-2-critical	Max $\|E\|$	Bound $\lfloor n^2/4 \rfloor$	Violators	Seconds
3	8	3	2	2	0	0.0001
4	64	7	4	4	0	0.0005
5	1024	27	6	6	0	0.0190
6	32768	571	9	9	0	0.5447
7	2097152	8883	12	12	0	43.4453

All rows pass Murty–Simon for labeled graphs on $n \leq 7$.

Note: counts are of **labeled** graphs (isomorphs counted separately). The inequality concerns edge cardinality, so labeled enumeration is a valid bound check; it is not an unlabeled census.

Balanced bipartite witnesses

For each $n = 3..7$, $K_{\lfloor n/2 \rfloor, \lceil n/2 \rceil}$ is diameter-2-critical, has exactly $\lfloor n^2/4 \rfloor$ edges, and attains the bound (machine-checked in the same script).

Artifact

`/workspace/mission-control/labs/erdos_742_murty_simon/results.json`
`/workspace/mission-control/experiments/band1/erdos_742/`

What this does not prove

- It does **not** prove Murty–Simon for all n (Fan–Füredi gap remains).
- It does **not** close `erdos_742`'s `answer(sorry)` in Lean.
- It does **not** re-verify Fan's $n \leq 24$ / $n = 26$ cases (out of tonight's wall budget).
- It does **not** claim any other Erdős problem.

Conclusion

One concrete Formal-Conjectures Erdős item: Problem **742** (Murty–Simon), stub present at `FormalConjectures/ErdosProblems/742.lean`.

Stub close/refute in Lean: blocked — Lean toolchain absent on box; documented above.

Finite check with honesty: exhaustive verification that every diameter-2-critical labeled graph on $n \leq 7$ vertices obeys $|E| \leq \lfloor n^2/4 \rfloor$, with equality attained by balanced complete bipartite graphs. Lean target stated for a future session: formalize `n  7` (or import Fan) without mass-claiming Erdős closures.

References

1. Formal Conjectures repository: google-deepmind/formal-conjectures, `FormalConjectures/ErdosProblems/742.lean`.
2. Erdős Problems site: https://www.erdosproblems.com/742
3. Fan, G. On diameter 2-critical graphs. *Discrete Math.* 67 (1987), 235–240.
4. Füredi, Z. The maximum number of edges in a minimal graph of diameter 2. *J. Graph Theory* 16 (1992), 81–98.
5. teorth/erdosproblems status taxonomy (decidable = open but reduced to finite computation).